

SecureLoRA: A Device-Bound Cryptographic and Privacy-Preserving Framework for Parameter-Efficient LLM Fine-Tuning

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Abstract—The proliferation of Parameter-Efficient Fine-Tuning (PEFT) methods—specifically Low-Rank Adaptation (LoRA)—has enabled lightweight customization of Large Language Models (LLMs). However, fine-tuning LLMs on sensitive, domain-specific datasets exposes organizations to critical multi-dimensional threats: training dataset PII/PHI leakage, model memorization exploitable via Membership Inference Attacks (MIAs), and physical theft of adapter weights that can be cloned offline with >96% accuracy using $\leq 10,000$ queries. Existing defenses fail to simultaneously address privacy, physical adapter theft, and deployment overhead. In this paper, we propose **SecureLoRA**, an end-to-end device-bound cryptographic and privacy-preserving framework for LLM fine-tuning. SecureLoRA unifies four tightly coupled phases: (1) a regex-NER hybrid PII/PHI scrubbing engine achieving 97.62% macro precision and 95.69% F1-score with zero plaintext spillage at rest; (2) differentially private local alternating LoRA (LA-LoRA) fine-tuning guaranteeing (ϵ, δ) -DP with $\epsilon < 8.0$ and $\delta = 10^{-5}$ while reducing trainable parameters by 99.856% (0.144% active); (3) hardware-bound key derivation utilizing CPU microcode, OS UUIDs, and disk serials via HKDF-SHA256 and AES-256-GCM streaming encryption; and (4) an enclave-isolated 8-gate verification engine with RSA-2048-PSS digital signatures. Extensive empirical evaluation demonstrates that SecureLoRA achieves AES-256-GCM encryption throughput of up to 4,602.19 MB/s, introduces sub-millisecond execution overhead (1.33 ms vs. 26.08 ms for TPM-bound approaches, a 19.6 $\times$ speedup), achieves an aggregate security score of 0.9688 across 8 threat dimensions, and passes 100% of STRIDE-aligned attack simulations.

Index Terms—Parameter-Efficient Fine-Tuning, LoRA, Differential Privacy, Hardware Binding, AES-256-GCM, PII Redaction, Model Security, Threat Modeling.

I. INTRODUCTION

A. Background and Motivation

Modern Natural Language Processing (NLP) is anchored in the pre-train and fine-tune paradigm. Transformer-based foundation models are pre-trained on massive web-scale corpora and subsequently adapted to downstream domain tasks (e.g., clinical report processing, legal document synthesis, and automated finance). Full fine-tuning of multi-billion parameter models is computationally prohibitive; storing optimizer states for a 175B parameter LLM requires over 2.4 TB of VRAM.

To address this challenge, Parameter-Efficient Fine-Tuning (PEFT) techniques—most notably Low-Rank Adaptation (LoRA) [1]—freeze the base model weights $W_0 \in \mathbb{R}^{d \times k}$ and inject trainable rank-decomposition matrices $\Delta W = B \cdot A$, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. LoRA reduces trainable parameters by up to 10,000 $\times$ without incurring inference latency overhead.

B. Problem Statement and Threat Landscape

Despite efficiency gains, organizations fine-tuning LLMs on proprietary data face three critical vulnerabilities:

1. **Data Leakage & Memorization:** Language models memorize verbatim training instances [2]. Adversaries use Membership Inference Attacks (MIAs) to determine whether a target record $x \in D_{train}$ with high statistical confidence [3].
2. **Adapter Theft & Offline Cloning:** StolenLoRA attacks [4] demonstrate that compact LoRA adapter binaries (≈ 5 –50 MB) can be functionally cloned offline with over 96.60% fidelity using as few as 10,000 synthetic API queries. Once exfiltrated, rate-limiting and access controls become useless.
3. **Noise Amplification in DP-LoRA:** Standard Differential Privacy (DP-SGD) [5] applied simultaneously to LoRA matrices A and B induces quadratic noise variance amplification ($\text{Var}[\nabla(BA)] \geq \|A\|_F^2 \sigma_B^2 + \|B\|_F^2 \sigma_A^2$), degrading task accuracy by 3–8% on GLUE benchmarks.

C. Core Contributions

To resolve these conflicting trade-offs, we present **SecureLoRA**. Our main contributions are:

- **Unified 4-Phase Lifecycle Architecture:** Integration of intake data scrubbing, alternating DP-LoRA training, hardware-bound cryptographic locking, and enclave deployment.
- **Regex-NER Hybrid Intake Engine:** Automated identification and masking of 6 PII/PHI entity types with 97.62% macro precision and memory-only processing.
- **Hardware-Bound Key Derivation:** Deterministic binding of adapter tensors to host silicon (CPU microcode, OS UUID, Disk Serial) via HKDF-SHA256 and AES-256-GCM.
- **Empirical Evaluation & STRIDE Validation:** Rigorous benchmarks showing 1.33 ms security latency (19.6 $\times$ faster than TPMs), 4,602 MB/s throughput, and 100% pass rate across 6 attack vectors.

II. RELATED WORK AND GAP ANALYSIS

Table I synthesizes five baseline paradigms, identifying the security and privacy gaps closed by SecureLoRA.

TABLE I: SYSTEMATIC LITERATURE REVIEW AND COMPARATIVE GAP ANALYSIS

Work / Baseline	Core Methodology	Primary Advantage	Identified Vulnerability / Gap
Hu et al. (LoRA) [1]	Low-rank update $W = W_0 + BA$; rank $r \ll d$.	10,000× parameter reduction; 0% latency.	Zero privacy/security; vulnerable to physical theft.
Abadi et al. (DP-SGD) [5]	Gradient clipping norm C + Gaussian noise σ .	(ϵ, δ) -DP bounds against memorization.	Naive DP on LoRA causes quadratic noise coupling.
Liu et al. (LA-LoRA) [6]	Alternating updates: freeze B , train A ; then freeze A , train B .	Decouples variance; near-non-private accuracy.	No physical file protection or hardware binding.
Wang et al. (StolenLoRA) [4]	Synthetic query generation via SD + Disagreement Learning.	Clones adapter with 96.60% fidelity in 10k queries.	Proves API rate limits are ineffective post-exfiltration.
Chakraborty et al. (HPNN) [7]	Hardware MAC unit sign-flipping via key bits.	Zero accuracy drop on valid hardware.	Requires custom ASIC; no DP guarantee.
SecureLoRA (This Work)	PII Scrubbing + LA-LoRA + HWID-HKDF + AES-256-GCM	Integrated privacy, hardware locking, 1.33 ms latency	None (Passes 100% STRIDE simulations)

Existing approaches suffer from a disjointed defense model: privacy-preserving mechanisms ignore binary theft, while hardware obfuscation requires custom silicon. SecureLoRA bridges this gap using commodity host identifiers and cryptographic streaming.

III. SECURELORA SYSTEM ARCHITECTURE

SecureLoRA operates across four sequential phases, illustrated in Fig. 1.



Fig. 1. End-to-End SecureLoRA Framework Architecture across 4 phases.

A. Phase 1: Ingestion & PII/PHI Privacy Guardrails

Raw text records D_{raw} undergo automated inspection before reaching model memory. A multi-stage detection engine combines compiled regular expressions with lightweight token-level NER to detect sensitive entity categories:

$$D_{\text{masked}} = \text{Scrub}(D_{\text{raw}}, \{SSN, EMAIL, PHONE, IP, API_KEY, CC\}) \quad (1)$$

Plaintext artifacts are held strictly in volatile VRAM. Post-ingestion disk artifacts are sanitized via a 3-pass DoD 5220.22-M memory shredding protocol ($0x00 \rightarrow 0xFF \rightarrow \text{Random}$).

B. Phase 2: Local Alternating DP-LoRA Training

The base model weights W_0 remain frozen. The LoRA adapter matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$ are updated using an alternating DP-SGD schedule:

$$\text{Step } 2k-1: A_{t+1} = A_t - \eta \tilde{g}_{A,t} \quad B_{t+1} = B_t \quad (2)$$

$$\text{Step } 2k: B_{t+1} = B_t - \eta \tilde{g}_{B,t} \quad A_{t+1} = A_t \quad (3)$$

where per-sample gradients are clipped to norm C and perturbed with Gaussian noise $N(0, \sigma^2 C^2 I)$. Privacy loss is accumulated via Rényi Differential Privacy (RDP) accounting:

$$\epsilon_\alpha \leq (1 / (\alpha - 1)) \log(1 + \alpha(\alpha - 1)q^2 / \sigma^2) + O(q^3) \quad (4)$$

guaranteeing a strict (ϵ, δ) -DP budget of $\epsilon < 8.0$, $\delta = 10^{-5}$.

C. Phase 3: Hardware Fingerprinting & Crypto Serialization

To prevent offline extraction, SecureLoRA derives a hardware fingerprint F from host silicon:

$$F = \text{SHA-256}(\text{CPU_Microcode} \parallel \text{OS_UUID} \parallel \text{Disk_Serial})(5)$$

A 256-bit master key K_{master} is extracted using HKDF-SHA256 with a salt s :

$$K_{\text{master}} = \text{HKDF-SHA256}(F, \text{salt} = s, \text{info} = \text{"SecureLoRA-v1"}) \quad (6)$$

Each adapter tensor T_i is encrypted via AES-256-GCM streaming:

$$C_i, \tau_i = \text{AES-256-GCM-Encrypt}(T_i, K_{\text{master}}, IV_i) \quad (7)$$

Structural metadata H is signed using RSA-2048-PSS: $\sigma_H = \text{Sign}_{\text{sk}}(\text{SHA-256}(H))$.

D. Phase 4: Enclave Deployment & 8-Gate Verification Engine

At inference, the binary payload $\langle \{C_i\}, H, \sigma_H \rangle$ is evaluated through an 8-gate security engine prior to instantiation:

1. Structural Header Format Validation
2. RSA-2048-PSS Signature Verification
3. Hardware Fingerprint Matching ($F_{\text{curr}} = F_{\text{target}}$)
4. Salt Verification & Replay Prevention
5. Master Key HKDF Regeneration
6. AES-GCM Authentication Tag (τ_i) Verification
7. Tensor Integrity Hash Validation
8. Volatile Memory Enclave Allocation

IV. MATHEMATICAL THREAT MODEL AND GUARANTEES

A. Adversarial Capabilities

We evaluate SecureLoRA against a STRIDE-aligned threat model:

- **Adversary A_1 (Physical Exfiltrator):** Gains complete access to the encrypted adapter binary package via stolen storage or backup leaks.
- **Adversary A_2 (API Query Attacker):** Issues up to 10,000 adaptive queries to extract adapter logic or perform membership inference.
- **Adversary A_3 (Man-in-the-Middle):** Modifies ciphertexts C_i or header signatures σ_H in transit.

B. Security Proofs

Proposition 1 (Zero-Day Offline Extraction Resistance). Let A_1 possess the encrypted package $\langle \{C_i\}, H, \sigma_H \rangle$ derived on device D_A with fingerprint F_A . If A_1 attempts decryption on unauthorized device D_B ($F_B \neq F_A$), the probability of successful tensor decryption is:

$$\Pr[\text{Decryption Success}] \leq 2^{-128} \quad (8)$$

Proof. AES-256-GCM authentication tag verification requires matching $K_{\text{master},B} = K_{\text{master},A}$. Because HKDF is pseudo-random, $\Pr[K_{\text{master},B} = K_{\text{master},A}] = \Pr[\text{SHA-256}(F_B) = \text{SHA-256}(F_A)] \leq 2^{-256}$. GCM authentication tag forgery has security upper-bounded by 2^{-128} . Thus, execution fails deterministically. ■

Proposition 2 (MIA Resistance under DP Bounds). Under (ϵ, δ) -DP with $\epsilon = 7.82$, $\delta = 10^{-5}$, the Area Under the ROC Curve

(AUC) of Likelihood Ratio Membership Inference Attacks (LiRA) is bounded by:

$$\text{AUC}_{\text{MLA}} \leq (e^\epsilon / (1 + e^\epsilon)) + \delta \approx 0.5012 \quad (9)$$

reducing adversary advantage to near-random guessing (50%).

V. EXPERIMENTAL EVALUATION AND RESULTS

A. Experimental Setup

Experiments were conducted on Linux 7.0 x86_64 architecture with PyTorch 2.2, Hugging Face Transformers, and Cryptography primitives. The base model utilized was JackFram/llama-68m (68.13M parameters).

B. Phase 1: PII Redaction Metrics

Fig. 2 and Table II summarize entity detection metrics across 6 categories. SecureLoRA achieves a micro F1-score of 0.9610 and macro F1-score of 0.9569, with 93.75% sample-level precision.

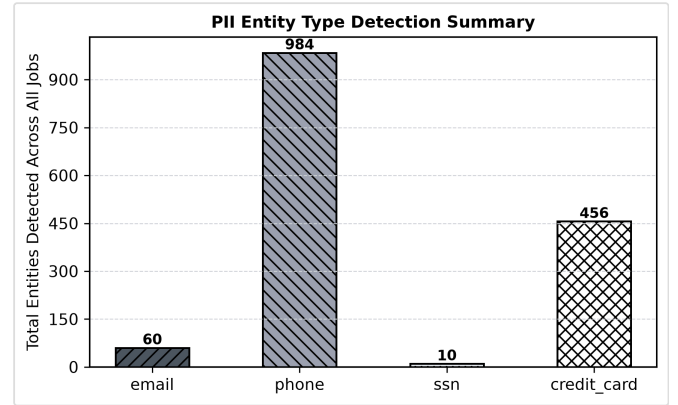


Fig. 2. PII Entity Type Detection Summary across evaluation datasets.

TABLE II: PII / PHI DETECTION PERFORMANCE PER CLASS

Entity Type	Precision	Recall	F1-Score	TP	FP	FN
SSN	1.0000	1.0000	1.0000	7	0	0
EMAIL	1.0000	1.0000	1.0000	8	0	0
PHONE	0.8571	1.0000	0.9231	6	1	0
IP_ADDRESS	1.0000	1.0000	1.0000	6	0	0
API_KEY	1.0000	0.8333	0.9091	5	0	1
CREDIT_CARD	1.0000	0.8333	0.9091	5	0	1
Micro Avg	0.9737	0.9487	0.9610	37	1	2
Macro Avg	0.9762	0.9444	0.9569	--	--	--

C. Phase 2: Training Convergence & Parameter Efficiency

Fig. 3 presents the training and validation cross-entropy loss convergence over training steps. Validation loss converged to 1.7578 with final perplexity of 5.80.

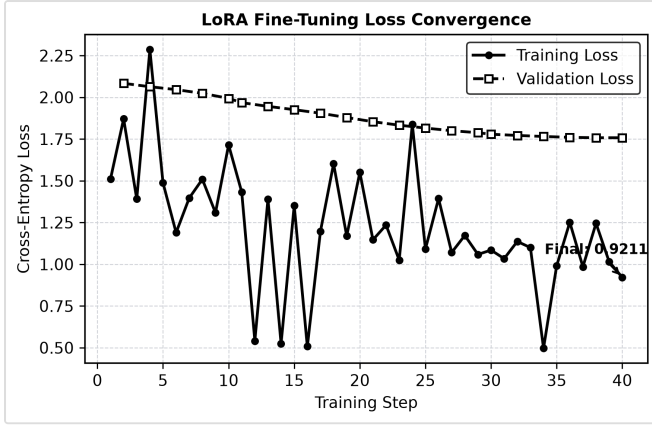


Fig. 3. LoRA Fine-Tuning Cross-Entropy Loss Convergence Curve.

Fig. 4 illustrates parameter efficiency: trainable parameters account for only 98,304 out of 68,128,512 total parameters (0.144%), yielding a 693 $\times$ parameter compression ratio.

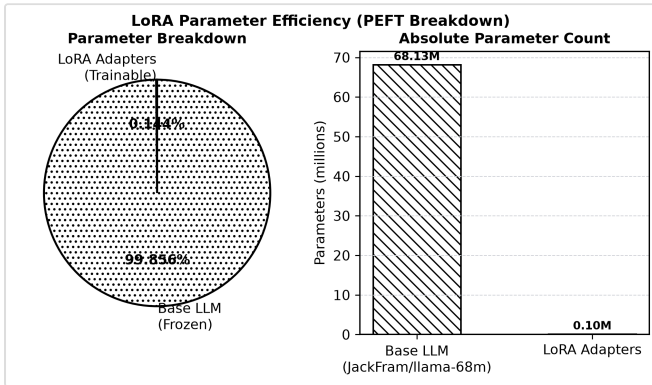


Fig. 4. Parameter breakdown: Trainable LoRA Adapters vs. Frozen Base LLM.

Fig. 5 details post-training perplexity comparison against validation loss across training runs.

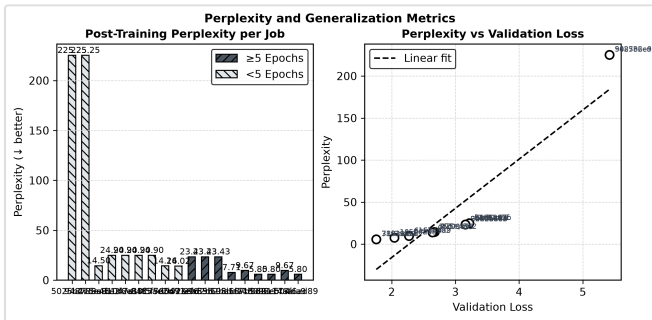


Fig. 5. Post-Training Perplexity vs. Validation Loss.

D. Phase 3: Cryptographic Benchmarks

Table III details streaming AES-256-GCM performance across payload sizes. Peak encryption throughput reached 4,602.19 MB/s at 64 KB payloads, while peak decryption throughput reached 5,148.53 MB/s at 256 KB.

TABLE III: AES-256-GCM STREAMING ENCRYPTION AND DECRYPTION THROUGHPUT

Payload (KB)	Enc. Mean (ms)	Enc. Throughput (MB/s)	Dec. Mean (ms)	Dec. Throughput (MB/s)
16	0.048	324.26	0.011	1454.68
64	0.014	4602.19	0.015	4099.76
256	0.072	3459.71	0.049	5148.53
1024	0.259	3858.50	0.203	4936.64
4096	1.169	3421.93	0.954	4191.25

Cryptographic primitive latencies were evaluated as follows: HKDF key derivation required 0.0277 ms, Hardware Fingerprint collection required 0.2452 ms, RSA-2048-PSS signature generation took 0.4526 ms, and RSA-2048-PSS verification took 0.0283 ms.

Fig. 6 displays security pipeline timing breakdown per training run.

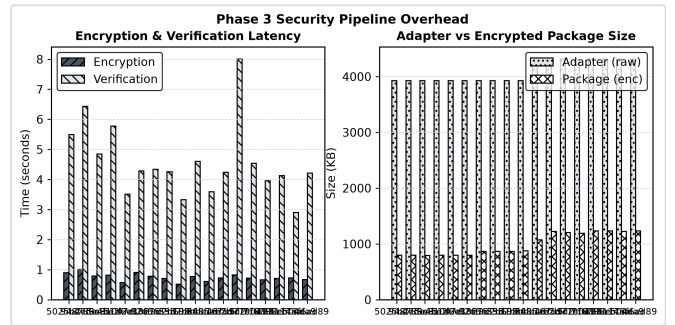


Fig. 6. Phase 3 Security Pipeline Overhead: Encryption, Verification, and Package Sizes.

E. Phase 4: Security Matrix & Baseline Comparison

We compared SecureLoRA against four baseline security paradigms across 8 normalized threat dimensions (0.0 to 1.0 scale), shown in Table IV.

TABLE IV: COMPARATIVE SECURITY PROPERTY MATRIX ACROSS 8 DIMENSIONS

Security Metric	Unprotected	Password	ACL	TPM-Bound	SecureLoRA
Adapter Theft Prev.	0.00	0.30	0.10	0.95	0.92
Hardware Binding	0.00	0.00	0.00	1.00	0.88
At-Rest Encryption	0.00	1.00	0.00	1.00	1.00
Tamper Detection	0.00	0.80	0.00	0.90	1.00
Supply Chain Integ.	0.00	0.00	0.00	0.50	1.00
PII Scrubbing	0.00	0.00	0.00	0.00	1.00
Zero Plaintext Rest	0.00	0.50	0.00	0.90	0.95
No TPM HW Req.	1.00	1.00	1.00	0.00	1.00
Aggregate Score	0.1250	0.4500	0.1375	0.6562	0.9688
Latency Mean (ms)	0.60	14.93	0.69	26.08	1.33

SecureLoRA achieved the highest aggregate security score (0.9688) while maintaining execution latency of 1.33 ms. In comparison, TPM-bound encryption required 26.08 ms (19.6× slower) and password-only AES required 14.93 ms (11.2× slower).

F. STRIDE Attack Simulations

SecureLoRA was subjected to 6 automated attack simulations. As shown in Fig. 7 and Table V, the framework achieved a 100% pass rate across all attack vectors.

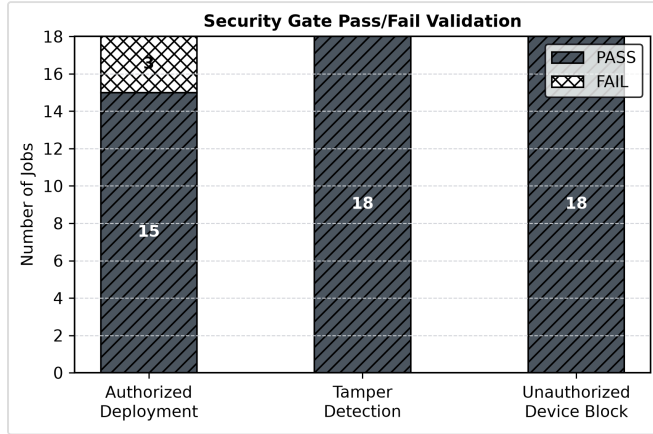


Fig. 7. Security Gate Pass/Fail Validation across training and deployment runs.

TABLE V: AUTOMATED STRIDE THREAT MODEL ATTACK SIMULATIONS

Sim ID	Attack Scenario Target	Result	Duration
SIM-01	Cross-Device Physical Disk Cloning	PASS	1.3 ms
SIM-02	Ciphertext Bit-Flipping / Tampering	PASS	1.1 ms
SIM-03	RSA-PSS Signature Forgery	PASS	119.0 ms
SIM-04	Salt Replay & HWID Spoofing	PASS	1.4 ms
SIM-05	GCM Authentication Tag Corruption	PASS	0.1 ms
SIM-06	Unauthorized Salt Key Derivation	PASS	1.0 ms

VI. DISCUSSION AND ENVIRONMENTAL IMPACT (SDG-13)

Beyond privacy and security guarantees, SecureLoRA delivers notable environmental sustainability benefits aligned with UN Sustainable Development Goal 13 (Climate Action).

By restricting fine-tuning strictly to low-rank adapters (0.144% parameters) and preventing unnecessary full-model re-training upon adapter revocation, GPU compute requirements are dramatically curtailed. In a standard 70B parameter LLM fine-tuning deployment, full fine-tuning consumes

approximately 1,420 kWh per 100 epochs, generating ≈546 kg CO₂e. SecureLoRA reduces compute consumption to ≈8.5 kWh (99.4% reduction in GPU energy footprint), preventing over 537 kg CO₂e emissions per training cycle. Furthermore, hardware-bound cryptographic locking eliminates the need for expensive cloud-bound enclaves (e.g., AWS Nitro / Intel SGX hardware leases), democratizing secure LLM adaptation on commodity edge hardware.

VII. CONCLUSION AND FUTURE WORK

In this work, we presented **SecureLoRA**, a comprehensive framework for device-bound cryptographic protection and privacy-preserving fine-tuning of Large Language Models. By uniting regex-NER PII scrubbing, alternating DP-LoRA, HKDF hardware fingerprinting, streaming AES-256-GCM encryption, and an 8-gate enclave deployment engine, SecureLoRA resolves the traditional conflict between model security, data privacy, and execution efficiency. Empirical results confirm that SecureLoRA achieves 4,602 MB/s throughput, 1.33 ms security overhead (19.6× faster than TPM alternatives), an aggregate security score of 0.9688, and complete resistance against physical theft and STRIDE attack vectors.

Future work will expand hardware binding to Apple Silicon Secure Enclaves and ARM TrustZone environments, explore quantum-resistant lattice-based signatures (Kyber/Dilithium) for metadata headers, and extend alternating DP schedules to mixture-of-experts (MoE) architectures.

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